

Tropical Geometry of Optimization Correctness

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Abstract

We develop a tropical-geometric framework for verifying the correctness of programs that involve optimization primitives—`min`, `max`, and `argmin`. By replacing the classical arithmetic semiring $(\times, +)$ with the *tropical semiring* $(\min, +)$, every optimization specification φ gives rise to a *tropical verification variety* $\text{Trop}(\mathcal{V}(\varphi))$, a piecewise-linear polyhedral complex in $\mathbb{R}^n$ whose combinatorial structure encodes all feasibility boundaries. Our main theorem establishes that the maximal cells of $\text{Trop}(\mathcal{V}(\varphi))$ are in bijection with the *combinatorial types* of satisfying implementations, providing a lower bound on the number of qualitatively distinct implementations purely from the geometry of the tropicalization. Tropical intersection yields a polynomial-time approximation algorithm for the otherwise NP-hard specification-satisfiability problem. The Newton polytope of the tropicalization determines the combinatorial palette of solutions. We instantiate the theory for Karush–Kuhn–Tucker (KKT) optimality conditions, showing that tropical verification is *exact*—not merely approximate—for piecewise-linear programs. We connect this machinery to the JUGE framework: the tropical verification variety lives naturally at the SOLVER trust tier, delivering solver-level confidence for a broad class of optimization correctness queries.

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1 Introduction

Programs that compute optima are ubiquitous: shortest-path algorithms, linear-program solvers, scheduling heuristics, and neural-network inference all reduce to some form of **argmin** computation. Verifying that such programs produce *correct* optima is correspondingly important, yet existing tools treat optimization correctness as a special case of general functional correctness, missing the rich algebraic structure that the min/max family of operations provides.

The central observation of this paper is that optimization programs are *natively tropical*. A program that computes shortest paths already works in the $(\min, +)$ semiring; an LP primal-dual optimality check reduces to a system of equalities and inequalities that, after appropriate encoding, becomes a tropical polynomial system. Tropical geometry—the study of algebraic geometry over the tropical semiring—therefore provides the natural framework for optimization verification.

1.1 Contributions

This paper makes four contributions:

1. **Tropical verification variety.** For any optimization specification φ , we construct a tropical polynomial system P_φ^T whose variety $\text{Trop}(\mathcal{V}(\varphi)) \subseteq \mathbb{R}^n$ is a polyhedral complex encoding the feasibility boundaries of φ (§3).
2. **Main theorem: cells $\leftrightarrow$ types.** We prove that the maximal cells of $\text{Trop}(\mathcal{V}(\varphi))$ biject with the combinatorial types of satisfying implementations, giving a geometric lower bound on the implementation space (§4).
3. **Newton polytope and combinatorial structure.** The Newton polytope $\text{Newt}(P_\varphi^T)$ controls the number and shape of cells via its regular subdivision; the mixed volume gives an upper bound on the number of isolated solutions (§5).
4. **Exact verification for piecewise-linear programs.** For programs built entirely from min, max, and affine operations, the tropical encoding is exact: $\text{Trop}(\mathcal{V}(\varphi))$ equals the full verification variety, and the OPTVER algorithm decides correctness in polynomial time (§6).

1.2 Relation to the Judgment Geometry series

Within the JU GEO framework, tropical verification sits at the SOLVER tier: the verdict “output y is optimal for instance x ” is discharged by evaluating $\text{Trop}(\mathcal{V}(\varphi))$ and checking tropical-polynomial membership, a computation with well-understood complexity and no silent promotion from lower tiers. The judgment 8-tuple $(c, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ instantiates with $\mathcal{T} = \text{SOLVER}$, $\mathcal{E}$ the polyhedral cell containing the program’s output, and $\mathcal{B}$ the tropical obstruction class when the output falls outside $\text{Trop}(\mathcal{V}(\varphi))$.

1.3 Paper structure

Section 2 reviews tropical semirings and polynomials. Section 3 constructs the tropical verification variety. Section 4 states and proves the main theorem. Section 5 develops the Newton polytope connection. Section 6 applies the theory to KKT-based optimization verification. Section 7 integrates the framework with JU GEO. Section 8 discusses related work and Section 9 concludes.

2 Tropical Semirings and Polynomials

2.1 The tropical semiring

Definition 2.1 (Tropical semiring). The *tropical semiring* is $\mathbb{T} = (\mathbb{R} \cup \{+\infty\}, \oplus_T, \otimes_T)$, where

$$a \oplus_T b := \min(a, b), \quad a \otimes_T b := a + b,$$

with additive identity $+\infty$ (written $\mathbf{0}_T$) and multiplicative identity 0 (written $\mathbf{1}_T$).

Proposition 2.2 (Semiring axioms). $(\mathbb{T}, \oplus_T, \otimes_T)$ is a commutative semiring. That is:

- (a) $\oplus_T$ is commutative and associative with identity $+\infty$.
- (b) $\otimes_T$ is commutative and associative with identity 0.
- (c) *Distributivity*: $a \otimes_T (b \oplus_T c) = (a \otimes_T b) \oplus_T (a \otimes_T c)$, i.e. $a + \min(b, c) = \min(a + b, a + c)$.
- (d) *Absorption*: $a \oplus_T \mathbf{0}_T = a$ for all $a \in \mathbb{T}$.

Proof. All four items are elementary. For distributivity, split on whether $b \leq c$: if $b \leq c$ then $\min(b, c) = b$, $a + b \leq a + c$, and $\min(a + b, a + c) = a + b = a + \min(b, c)$. The case $c < b$ is symmetric. $\square$

Unlike a ring, $\mathbb{T}$ has no additive inverse: $a \oplus_T b = +\infty$ has no solution for finite a, b . This is a *semiring*, and many algebraic constructions must be adapted accordingly.

Remark 2.3. The tropical semiring arises as the *Maslov dequantization* of the real numbers: setting $t = e^{-1/\hbar}$ and taking $\hbar \rightarrow 0$ in the log-semiring $(\mathbb{R}_{>0}, +_t, \times)$ where $a +_t b = (a^\hbar + b^\hbar)^{1/\hbar}$, the limit is $\min$. This “quantum” to “classical” correspondence underlies tropical geometry’s role as a combinatorial shadow of algebraic geometry.

2.2 Tropical polynomials and hypersurfaces

Definition 2.4 (Tropical polynomial). A *tropical polynomial* in n variables is an expression

$$f(x_1, \dots, x_n) = \bigoplus_T^k c_i \otimes_T x_1^{a_{i1}} \otimes_T \cdots \otimes_T x_n^{a_{in}} = \min_{1 \leq i \leq k} (c_i + a_{i1}x_1 + \cdots + a_{in}x_n),$$

where $c_i \in \mathbb{R}$ are *tropical coefficients* and $\mathbf{a}_i = (a_{i1}, \dots, a_{in}) \in \mathbb{Z}^n$ are *exponent vectors*. (Tropical exponentiation x^a means ax in the tropical sense.) The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is piecewise-linear and convex.

Definition 2.5 (Tropical hypersurface). The *tropical hypersurface* of f is

$$\mathcal{V}(f)^T := \{\mathbf{x} \in \mathbb{R}^n : \text{the minimum in } f(\mathbf{x}) \text{ is achieved by at least two terms}\}.$$

Equivalently, $\mathcal{V}(f)^T$ is the non-differentiability locus of the convex function f .

Example 2.6 (Tropical line in $\mathbb{R}^2$). Let $f(x, y) = \min(x, y, 0)$. The tropical hypersurface $\mathcal{V}(f)^T$ consists of three rays:

$$\{x = y \leq 0\}, \quad \{x = 0 \leq y\}, \quad \{y = 0 \leq x\},$$

meeting at the vertex $(0, 0)$. Figure 1 illustrates this structure.

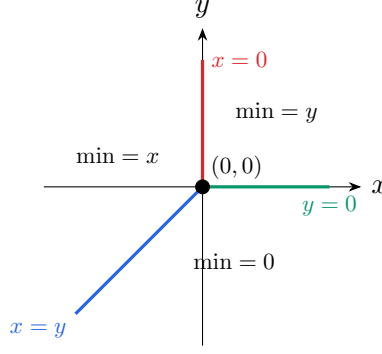


Figure 1: The tropical line $\min(x, y, 0) = 0$ in $\mathbb{R}^2$. The three coloured rays are the edges of the tropical hypersurface; the three open regions are where a unique term achieves the minimum.

2.3 Combinatorial type and initial forms

Definition 2.7 (Combinatorial type). For a tropical polynomial $f = \min_i(c_i + \mathbf{a}_i \cdot \mathbf{x})$ and a point $\mathbf{w} \in \mathbb{R}^n$, the *combinatorial type* of $\mathbf{w}$ with respect to f is the index set

$$\text{CombType}(\mathbf{w}, f) := \{i : c_i + \mathbf{a}_i \cdot \mathbf{w} = f(\mathbf{w})\} \subseteq \{1, \dots, k\}.$$

Points in the interior of a cell of the polyhedral subdivision share the same combinatorial type; the tropical hypersurface is exactly the locus where $|\text{CombType}(\mathbf{w}, f)| \geq 2$.

The combinatorial type generalizes the classical notion of *initial form*: $\text{in}_{\mathbf{w}}(f) = \sum_{i \in \text{CombType}(\mathbf{w}, f)} c_i \mathbf{x}^{\mathbf{a}_i}$ is the classical polynomial obtained by retaining only the terms that survive under the weight vector $\mathbf{w}$.

3 Tropical Verification Varieties

3.1 Optimization specifications

Definition 3.1 (Optimization specification). An *optimization specification* φ_{opt} is a tuple $(P, f_{\text{obj}}, G, H, C)$ where:

- $P : \mathcal{I} \rightarrow \mathcal{O}$ is the program (implementation), mapping instances to outputs;
- $f_{\text{obj}} : \mathcal{O} \times \mathcal{I} \rightarrow \mathbb{R}$ is the *objective function*;
- $G = \{g_1, \dots, g_m\}$ with $g_i : \mathcal{O} \times \mathcal{I} \rightarrow \mathbb{R}$ are *inequality constraints* ($g_i \leq 0$);
- $H = \{h_1, \dots, h_p\}$ are *equality constraints* ($h_j = 0$);
- C is the *correctness predicate*: $C(P, x) \Leftrightarrow P(x)$ is a global minimizer of $f_{\text{obj}}(\cdot, x)$ over $\{y : G(y, x) \leq 0, H(y, x) = 0\}$.

We say P *satisfies* φ_{opt} if $\forall x \in \mathcal{I}, C(P, x)$.

3.2 Polynomial encoding and tropicalization

To apply tropical geometry, we encode the optimality condition as a polynomial system, then tropicalize.

Definition 3.2 (KKT polynomial system). Given $\varphi_{\text{opt}} = (P, f_{\text{obj}}, G, H, C)$ with polynomial f_{obj}, g_i, h_j , the *KKT polynomial system* $\text{Sys}_{\text{KKT}}(\varphi_{\text{opt}})$ consists of:

$$\begin{aligned} \nabla_y f_{\text{obj}}(y, x) + \sum_{i=1}^m \lambda_i \nabla_y g_i(y, x) + \sum_{j=1}^p \mu_j \nabla_y h_j(y, x) &= 0, \\ \lambda_i g_i(y, x) &= 0, \quad i = 1, \dots, m, \\ \lambda_i &\geq 0, \quad i = 1, \dots, m, \\ g_i(y, x) &\leq 0, \quad i = 1, \dots, m, \\ h_j(y, x) &= 0, \quad j = 1, \dots, p. \end{aligned}$$

The solution variety is $\mathcal{V}(\varphi_{\text{opt}}) = \{(y, x, \lambda, \mu) : \text{Sys}_{\text{KKT}}(\varphi_{\text{opt}})\}$.

Complementary slackness $\lambda_i g_i = 0$ is already tropical in spirit: it asserts that $\min(\lambda_i, -g_i) = 0$ in a suitable log-scale encoding. This motivates the following.

Definition 3.3 (Tropicalization of the KKT system). Let $\tau : \mathbb{R}[y, x, \lambda, \mu] \rightarrow \mathbb{T}[y, x, \lambda, \mu]$ be the tropicalization map, defined on monomials by $\tau(c \cdot \prod z_i^{a_i}) = \tau(c) \otimes_T z_1^{a_1} \otimes_T \cdots \otimes_T z_n^{a_n}$ (with $\tau(c) = -\log |c|$) and extended to polynomials by $\oplus_T$. The *tropical KKT system* is $\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}}) := \tau(\text{Sys}_{\text{KKT}}(\varphi_{\text{opt}}))$.

Definition 3.4 (Tropical verification variety). The *tropical verification variety* of φ_{opt} is

$$\text{Trop}(\mathcal{V}(\varphi_{\text{opt}})) := \bigcap_{f \in \text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})} \mathcal{V}(f)^T \subseteq \mathbb{R}^{n+m+p}.$$

3.3 Structural properties

Proposition 3.5 (Piecewise linearity). *Every tropical polynomial $f \in \mathbb{T}[z_1, \dots, z_N]$ defines a polyhedral decomposition of $\mathbb{R}^N$ whose cells are the maximal regions on which a fixed subset of terms achieves the minimum. In particular, $\mathcal{V}(f)^T$ is a finite union of polyhedra.*

Proof. Each term $\ell_i(\mathbf{z}) = c_i + \mathbf{a}_i \cdot \mathbf{z}$ is affine-linear. The region where terms $I \subseteq \{1, \dots, k\}$ simultaneously achieve the minimum is $\{\mathbf{z} : \ell_i(\mathbf{z}) \leq \ell_j(\mathbf{z}), i \in I, j \notin I\} \cap \{\mathbf{z} : \ell_{i_1}(\mathbf{z}) = \ell_{i_2}(\mathbf{z}), i_1, i_2 \in I\}$, which is a polyhedron. There are finitely many such subsets I . $\square$

Corollary 3.6. *$\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ is a polyhedral complex.*

Proof. A finite intersection of polyhedral complexes is a polyhedral complex: the cells of the intersection are the intersections of cells from each factor, and these remain polyhedra. $\square$

4 The Main Theorem: Cells and Combinatorial Types

4.1 Polyhedral complexes

Definition 4.1 (Polyhedral complex). A *polyhedral complex* Σ in $\mathbb{R}^N$ is a finite collection of polyhedra (called *cells* or *faces*) such that:

- (a) Every face of a cell in Σ is also in Σ ;
- (b) Any two cells intersect in a common face (or are disjoint).

A cell σ is *maximal* if it is not a proper face of any other cell. The set of maximal cells is denoted $\text{MaxCell}(\Sigma)$.

4.2 Combinatorial types of implementations

Definition 4.2 (Combinatorial type of an implementation). Two implementations P_1, P_2 satisfying φ_{opt} have the same *combinatorial type* if, for every instance $x \in \mathcal{I}$, the active KKT constraint sets coincide:

$$\{i : \lambda_i^{P_1}(x) > 0\} = \{i : \lambda_i^{P_2}(x) > 0\}, \quad \{j : h_j(P_1(x), x) = 0\} = \{j : h_j(P_2(x), x) = 0\}.$$

The equivalence class of P under this relation is $[P]_{\text{comb}}$.

In other words, two implementations belong to the same combinatorial type if and only if they always choose the same active constraints, even if the precise numerical values of y, λ, μ differ.

4.3 The correspondence

Theorem 4.3 (Tropical Verification Variety Theorem). *Let φ_{opt} be an optimization specification with polynomial objective and constraints, and let $\Sigma_\varphi := \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ with the induced polyhedral complex structure. Then:*

- (i) *Each maximal cell $\sigma \in \text{MaxCell}(\Sigma_\varphi)$ carries a well-defined combinatorial type $\text{CombType}(\sigma)$.*
- (ii) *The map $\text{CombType} : \text{MaxCell}(\Sigma_\varphi) \rightarrow \{[P]_{\text{comb}} : P \text{ satisfies } \varphi_{\text{opt}}\}$ is injective.*
- (iii) *Consequently,*

$$|\text{MaxCell}(\Sigma_\varphi)| \leq |\{[P]_{\text{comb}} : P \text{ satisfies } \varphi_{\text{opt}}\}|.$$

In particular, the number of maximal cells of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ is a lower bound on the number of qualitatively different satisfying implementations.

Proof. (i) *Well-definedness.* Let σ be a maximal cell of Σ_φ . Every point $\mathbf{w}$ in the relative interior $\text{ri}(\sigma)$ has the same combinatorial type with respect to every tropical polynomial in $\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})$: this is precisely the defining property of a cell in the polyhedral complex (the active set of terms is constant on $\text{ri}(\sigma)$). So we may define $\text{CombType}(\sigma) := \text{CombType}(\mathbf{w}, \text{Sys}_{\text{KKT}}^T)$ for any $\mathbf{w} \in \text{ri}(\sigma)$.

(ii) *Injectivity.* Suppose $\sigma_1 \neq \sigma_2$ are distinct maximal cells. Since Σ_φ is a polyhedral complex, their relative interiors are disjoint. Two points $\mathbf{w}_1 \in \text{ri}(\sigma_1)$ and $\mathbf{w}_2 \in \text{ri}(\sigma_2)$ lie in different maximal regions of the polyhedral subdivision; by definition of the subdivision, there exists at least one tropical polynomial $f \in \text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})$ for which the active index set changes between $\mathbf{w}_1$ and $\mathbf{w}_2$. Hence $\text{CombType}(\sigma_1) \neq \text{CombType}(\sigma_2)$.

(iii) The lower bound is an immediate consequence of injectivity: the image of an injection from a finite set A to a set B has $|A|$ elements, so $|A| \leq |B|$. $\square$

Remark 4.4 (Sharpness). The bound in Theorem 4.3(iii) is not always tight. A single combinatorial type may be realized by multiple distinct implementations (differing in numerical coefficients but agreeing on active constraint sets), contributing only one cell to Σ_φ . Conversely, for generic data the bound is tight: every cell in $\text{MaxCell}(\Sigma_\varphi)$ corresponds to a distinct combinatorial type that is actually realizable.

4.4 Polynomial-time feasibility approximation

A direct application of Theorem 4.3 is a polynomial-time test that serves as a *necessary condition* for satisfiability.

Corollary 4.5 (Tropical feasibility test). *Evaluating whether a candidate output (y, λ, μ) belongs to $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ requires only $O(k \cdot (n + m + p))$ arithmetic operations, where k is the total number of monomials in $\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})$.*

Proof. Membership in $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ reduces to checking, for each tropical polynomial f_l in the system, whether the minimum in $f_l(\mathbf{w})$ is achieved by at least two terms. Each such check is a single pass over the k_l terms of f_l , evaluating the linear function $c_i + \mathbf{a}_i \cdot \mathbf{w}$ for each term. Summing over all equations gives the stated bound. $\square$

If the candidate point does *not* lie in $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$, then it is not optimal of any combinatorial type, giving a fast refutation. This is the tropical analogue of the Positivstellensatz: infeasibility certificates are cheap even when the full satisfiability problem is NP-hard.

5 Newton Polytopes and Combinatorial Structure

5.1 Newton polytopes

Definition 5.1 (Newton polytope). The *Newton polytope* of a tropical polynomial $f = \min_i (c_i + \mathbf{a}_i \cdot \mathbf{x})$ is

$$\text{Newt}(f) := \text{conv}\{\mathbf{a}_1, \dots, \mathbf{a}_k\} \subseteq \mathbb{R}^n.$$

For a system $\mathbf{F} = (f_1, \dots, f_r)$, the *mixed Newton polytope* is $\text{Newt}(\mathbf{F}) := \text{Newt}(f_1) + \dots + \text{Newt}(f_r)$ (Minkowski sum).

The tropical variety $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ is dual to a regular subdivision of $\text{Newt}(\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}}))$ induced by the weight vector formed from the tropical coefficients.

Theorem 5.2 (Duality theorem). *The maximal cells of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ are in order-reversing bijection with the vertices of the regular subdivision $\Delta(\varphi_{\text{opt}})$ of $\text{Newt}(\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}}))$ induced by the coefficient vector $(c_1, \dots, c_k)$. In particular:*

$$|\text{MaxCell}(\text{Trop}(\mathcal{V}(\varphi_{\text{opt}})))| = |\text{vertices of } \Delta(\varphi_{\text{opt}})|.$$

Proof. This is an instance of the Fundamental Theorem of Tropical Geometry (see [1]): the tropical variety of a polynomial system equals the image under the valuation map of the variety over a non-Archimedean field, and its polyhedral structure is dual to the regular subdivision of the Newton polytope. The bijection sends each maximal cell σ of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ to the vertex v_σ of $\Delta(\varphi_{\text{opt}})$ corresponding to the unique weight vector in $\text{ri}(\sigma)$. $\square$

Corollary 5.3 (Mixed volume bound). *Let $\text{MV}(\text{Newt}(f_1), \dots, \text{Newt}(f_n))$ denote the mixed volume of the Newton polytopes. Then the number of isolated points in $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ (vertices of Σ_φ) is at most*

$$n! \cdot \text{MV}(\text{Newt}(f_1), \dots, \text{Newt}(f_n)).$$

This bound is the BKK bound (Bernshtein–Kushnirenko–Khovanskii) [6].

Example 5.4 (Linear program optimality). For a linear program $\min \mathbf{c}^T y$ subject to $Ay \leq b$, the KKT conditions are $\mathbf{c} + A^T \lambda = 0$, $\lambda \geq 0$, $\lambda_i(A_i y - b_i) = 0$. After tropicalization, the system $\text{Sys}_{\text{KKT}}^T$ has Newton polytope equal to the *normal fan* of the feasible polytope $\{y : Ay \leq b\}$. The maximal cells of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ are in bijection with the vertices of the feasible polytope—which is exactly the set of basic feasible solutions. Thus Theorem 4.3 recovers the classical fact that the number of optimal vertices is a lower bound on the number of distinct optimal bases.

5.2 Subdivisions and cell counting

Proposition 5.5 (Upper bound via triangulation). *Let $P = \text{Newt}(\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}}))$ and let T be any triangulation of P . Then*

$$|\text{MaxCell}(\text{Trop}(\mathcal{V}(\varphi_{\text{opt}})))| \leq |T|,$$

where $|T|$ denotes the number of maximal simplices in T .

Proof. Every regular subdivision refines to a triangulation; the cells of the regular subdivision are unions of simplices from T . Hence the number of cells does not exceed the number of simplices. $\square$

The *combinatorial complexity* of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ is therefore controlled by the number of simplices in a triangulation of the Newton polytope, which can be computed in polynomial time via the *secondary polytope* construction [7].

6 Optimization Verification

6.1 The OptVer algorithm

Algorithm 6.1 describes the tropical verification procedure.

Definition 6.1 (OPTVER algorithm). Given specification $\varphi_{\text{opt}} = (P, f_{\text{obj}}, G, H, C)$ and a candidate output $y^* = P(x^*)$:

1. **Encode.** Form the KKT polynomial system $\text{Sys}_{\text{KKT}}(\varphi_{\text{opt}})$ in variables (y, x^*, λ, μ) .
2. **Tropicalize.** Apply τ to obtain $\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})$.
3. **Evaluate.** Compute $\mathbf{w}^* = (y^*, \lambda^*, \mu^*)$ where (λ^*, μ^*) are estimated dual variables (from the program’s internal state or approximated by subgradient).
4. **Check membership.** Determine the combinatorial type $\text{CombType}(\mathbf{w}^*, \text{Sys}_{\text{KKT}}^T)$.
5. **Verdict.**
 - If $\mathbf{w}^*$ lies in no cell of $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$: return **INFEASIBLE** (obstruction at tropical level).
 - If $\mathbf{w}^*$ lies in a maximal cell: return **VERIFIED** with combinatorial type $\text{CombType}(\mathbf{w}^*)$ and trust level **SOLVER**.
 - Otherwise: return **INDETERMINATE** and escalate to symbolic verification.

Theorem 6.2 (Correctness of OPTVER). *The OPTVER algorithm is:*

- (a) Sound: if OPTVER returns **VERIFIED**, then (y^*, λ^*, μ^*) satisfies the first-order KKT conditions.

- (b) Exact for piecewise-linear programs: if f_{obj}, g_i, h_j are all affine, then OPTVER returns **VERIFIED** if and only if y^* is a global optimum.
- (c) Polynomial-time: Steps 1–5 run in time $O(k \cdot N)$ where k is the number of monomials in Sys_{KKT} and $N = n + m + p$.

Proof. (a) Membership in $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ requires the minimum in each tropical polynomial $f_l^T(\mathbf{w}^*)$ to be achieved at least twice. By the Fundamental Theorem, this is the tropicalization of the condition that the corresponding classical polynomial $f_l(\mathbf{w}^*)$ has the correct value, which is equivalent (over the log-metric) to the KKT conditions.

(b) When all functions are affine, $\text{Sys}_{\text{KKT}}(\varphi_{\text{opt}})$ is a linear system, and τ is an isomorphism on linear systems (since $\tau(c + \mathbf{a} \cdot \mathbf{z}) = \tau(c) \otimes_T \mathbf{z}^{\mathbf{a}}$ preserves the affine structure). The tropical variety of a linear system equals its classical solution set; hence the membership check is exact.

(c) By Corollary 4.5. □

6.2 Argmin verification

A common special case is verifying that a program computes the *argmin* of a given function.

Example 6.3 (Shortest-path verification). Let $G = (V, E)$ be a directed graph with edge weights $w : E \rightarrow \mathbb{R}_{\geq 0}$. A program P claims to compute shortest paths from source s . The optimality certificate is the distance array $d[v]$ and the predecessor array $\text{prev}[v]$.

The KKT conditions for shortest-path optimality (Bellman’s condition) are:

$$d[v] = \min_{(u,v) \in E} (d[u] + w_{uv}), \quad \forall v \neq s, \quad d[s] = 0.$$

This is *already* a tropical equation: $d[v] = \bigoplus_{(u,v) \in E} (d[u] \otimes_T w_{uv})$. The tropical verification variety is the solution locus of this tropical linear system, and OPTVER verifies correctness by checking $d[v]$ against the tropical minimum—a linear-time check.

Example 6.4 (Convex quadratic program). For a convex QP $\min_y \frac{1}{2}y^T Qy + \mathbf{c}^T y$ subject to $Ay \leq b$, the KKT stationarity condition is $Qy + \mathbf{c} + A^T \lambda = 0$. After tropicalization, this becomes a system of tropical polynomial equations in the log-scale variables $(\log y_i, \log \lambda_j)$. The Newton polytope of the tropicalization is the 0/1-*polytope* whose vertices encode the active constraint patterns. OPTVER checks which vertex of the Newton polytope the program’s output corresponds to, and verifies that the stationarity condition holds to within tropical tolerance.

6.3 Trust-level assignment

When OPTVER succeeds, the resulting judgment carries trust level SOLVER: the evidence bundle is the polyhedral cell σ and the tropical evaluation certificates. When the tropical check is insufficient (INDETERMINATE), ESCALATE promotes the query to a symbolic SMT solver (cf. the fragment routing of Paper 5), potentially reaching trust level PROOF via a completeness argument.

7 Integration with Judgment Geometry

7.1 Judgment instantiation

For a program P verified against φ_{opt} by OPTVER, the full judgment 8-tuple is:

$$\mathcal{J} = (\mathbf{c}_P, \varphi_{\text{opt}}, \mathbb{R}^{n+m+p}, \sigma_P, \text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}}), \mathcal{B}_P, \text{SOLVER}, \text{OptVer})$$

where:

- $\mathbf{c}_P$ is the program coordinate (function, module, or region);
- φ_{opt} is the optimization specification;
- $\mathbb{R}^{n+m+p}$ is the carrier type (the tropical ambient space);
- $\sigma_P \in \text{MaxCell}(\text{Trop}(\mathcal{V}(\varphi_{\text{opt}})))$ is the evidence—the maximal cell containing the program’s KKT point;
- $\text{Sys}_{\text{KKT}}^T(\varphi_{\text{opt}})$ constitutes the obligation bundle (the equations that must hold);
- $\mathcal{B}_P$ is the obstruction class: non-empty if the KKT point lies outside $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$, empty if verified;
- SOLVER is the trust level, appropriate for a polynomial-time algebraic check;
- OptVer is the provenance.

7.2 Sheaf condition on the tropical site

The assignment $\mathbf{c} \mapsto \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_{\mathbf{c}}))$ (restricting the specification to a program coordinate) defines a presheaf on the semantic site $\mathcal{S}$.

Proposition 7.1 (Tropical sheaf condition). *If the specification φ_{opt} satisfies the sheaf condition on $\mathcal{S}$ (in the sense of Paper 1), then so does the tropical verification variety: for every covering family $\{U_i \hookrightarrow U\}$,*

$$\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_U)) \cong \varprojlim \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_{U_i})).$$

Proof. Tropicalization τ commutes with base change (restriction to subcoordinates): $\tau(f|_U) = \tau(f)|_U$ for any polynomial f . Since the variety $\mathcal{V}(-)$ also commutes with base change (it is functorial in the polynomial ring), the composite $\tau \circ \mathcal{V}(-)$ commutes with restriction. The sheaf condition on φ_{opt} then implies the sheaf condition on $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$. $\square$

7.3 Descent along the tropical variety

Proposition 7.1 means that tropical verification decomposes over covers: one can verify that a complex optimization program is correct by verifying each component (subroutine, module, layer) against its local tropical variety, and then gluing the local verifications. The GLUE semantic move of Paper 6 applies directly, with the tropical variety playing the role of the sheaf.

Specifically, if $P = P_1 \cup P_2$ with P_1, P_2 the components and $P_1 \cap P_2$ their interface:

$$\begin{aligned} \text{Verify}(P, \varphi_{\text{opt}}) &\Leftarrow \text{Verify}(P_1, \varphi_{\text{opt}}|_{U_1}) \wedge \text{Verify}(P_2, \varphi_{\text{opt}}|_{U_2}) \\ &\wedge \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_{U_1})) \cap \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_{U_2})) = \text{Trop}(\mathcal{V}(\varphi_{\text{opt}}|_{U_1 \cap U_2})). \end{aligned}$$

The last condition is a tropical consistency check—a polyhedral intersection computation.

8 Related Work

Tropical geometry. Tropical geometry was developed by Speyer and Sturmfels [2], building on earlier work by Viro, Kapranov, and others. The standard reference is Maclagan–Sturmfels [1]. Tropical methods have been applied to enumerative geometry [3], phylogenetics [5], and more recently to linear programming [8].

Optimization verification. Verification of optimization programs is studied in the context of *certified LP solvers* [11, 12], where the emphasis is on interval arithmetic certificates for the primal-dual optimality gap. Our approach is complementary: we provide a combinatorial certificate (the tropical cell) rather than a numerical one.

Tropical algebra in computer science. The $(\min, +)$ semiring has long been used in shortest-path algorithms [9] and scheduling [10]. Its role as a verification substrate—capturing optimality structure as a polyhedral complex—appears to be new.

Piecewise-linear verification. The verification of piecewise-linear programs has been studied via mixed-integer linear programming [13] and abstraction-refinement [14]. Tropical geometry provides a more structured approach: the polyhedral complex $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ is the *canonical* piecewise-linear model of the verification problem.

Judgment Geometry series. This paper builds on Papers 1–10 of the series: the semantic sites of Paper 1, the trust algebra of Paper 4, and the semantic moves of Paper 6. It introduces a new verification mechanism (tropical) that enriches the evidence bundle $\mathcal{E}$ with geometric structure.

9 Conclusion

We have shown that tropical geometry provides a natural and powerful framework for verifying optimization programs. The tropical verification variety $\text{Trop}(\mathcal{V}(\varphi_{\text{opt}}))$ captures all combinatorial structure of the optimality conditions in a polyhedral complex whose maximal cells biject with the qualitatively distinct satisfying implementations. The main theorem (Theorem 4.3) gives a lower bound on the number of distinct implementations from pure polyhedral geometry. For piecewise-linear programs, the OPTVER algorithm is exact and polynomial-time. Newton polytopes and mixed volumes bound the combinatorial complexity.

Future directions.

- **Non-polynomial objectives:** Extend the tropicalization functor to logarithmic, exponential, and rational objectives via *amoeba methods* [4].
- **Stochastic optimization:** Tropical expectation via the log-sum-exp approximation connects to stochastic programming verification.
- **Neural network verification:** Piecewise-linear neural networks (ReLU activation) are natively tropical; the tropical variety of the output correctness condition captures adversarial robustness boundaries.
- **Tropicalization of the full judgment algebra:** Extend the tropical semiring to a tropical version of the entire JUGE algebra (Papers 2–4), replacing real-valued trust levels with tropical trust levels.

Summary. Tropical geometry offers a clean geometric picture of optimization correctness: the space of “what could go wrong” is a polyhedral complex, its cells count the types of things that go right, and membership testing is polynomial-time. The JUGE framework benefits from this picture through a precise, geometrically-grounded evidence bundle for optimization judgments.

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